

My Product Manager Upload API

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Introduction

The purpose of this API is to give data suppliers a means to upload their product information to the 'My Product Manager' application of GS1 Belgium & Luxembourg. The product information within this system is made available to the data recipients according to the publication settings defined by the data supplier uploading the data.

Environments

GS1 provides two environments that can be connected to (UAT and PRD). All API calls during development can be executed against the UAT environment. The endpoints can be switched to the production environment when your application is promoted to production.

UAT:

- Authorization endpoint: <https://login-uat.gs1belu.org/oauth/token>
- API endpoint: [https://api-
uat.gs1belu.org/myproductmanager/upload/v6/tradeitems](https://api-uat.gs1belu.org/myproductmanager/upload/v6/tradeitems)

PROD:

- Authorization endpoint: <https://login.gs1belu.org/oauth/token>
- API endpoint: <https://api.gs1belu.org/myproductmanager/upload/v6/tradeitems>

PLEASE NOTE: the API endpoint contains the version of the JSON contract (e.g. V6) so you can define the used contract yourself. The data models are updated 4 times a year (according to the sector requirements) and GS1 tries to support at least the latest version of the API contract + 1 previous version. We advise data supplier to switch to the latest version of the API contract as soon as possible.

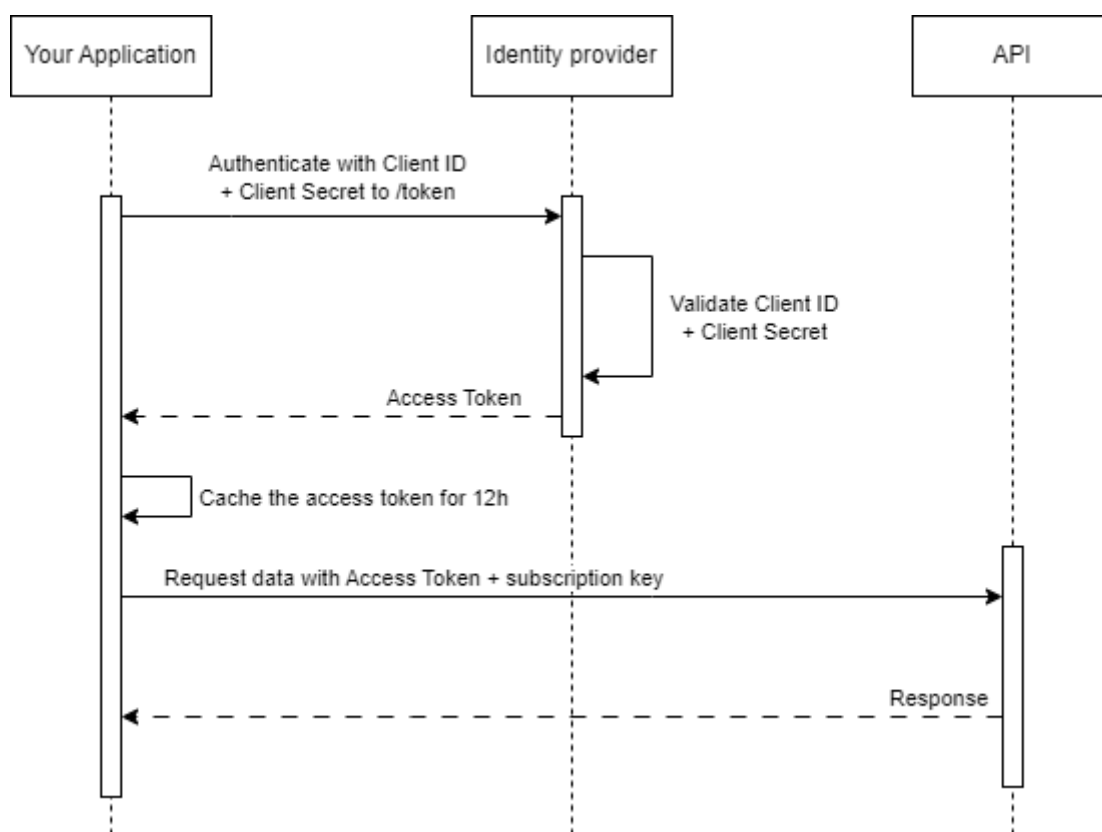
The latest contract information is always available at: [How can I create product data in My Product Manager? | GS1 Belgium & Luxembourg \(gs1belu.org\)](https://gs1belu.org/How-can-I-create-product-data-in-My-Product-Manager/)

Security

This API is secured by OAuth 2.0, specifically with the client credentials flow. This flow allows the application on your server to be authenticated and authorized with our identity provider.

The high-level flow is as follows:

1. Your app authenticates with the Authorization Server using its Client ID and Client Secret on the authorization endpoint.
2. The Authorization Server validates the Client ID and Client Secret.
3. The Authorization Server responds with an Access Token.
4. The Access Token is valid for 12 hours. **The token must be cached and re-used during that period. Users that don't respect this rule will be disconnected.**
5. Your application can use the Access Token to call the TradeItem API on behalf of itself.
6. The TradeItem API responds with requested data.



An example of the request to fetch the access token is shown below. Cache the value of "access_token" for 12 hours (please note the example below is for PROD)

```
▼ POST https://login.gs1belu.org/oauth/token
  ► Network
  ▼ Request Headers
    Content-Type: "application/x-www-form-urlencoded"
    User-Agent: "PostmanRuntime/7.23.0"
    Accept: "*/*"
    Cache-Control: "no-cache"
    Postman-Token: "e4a01d09-184b-477c-bdac-41ab23e80d3e"
    Host: "login.gs1belu.org"
    Accept-Encoding: "gzip, deflate, br"
    Content-Length: 193
    Connection: "keep-alive"
  ▼ Request Body
    client_id: "[REDACTED]"
    client_secret: "[REDACTED]"
    audience: "https://api.gs1belu.org/"
    grant_type: "client_credentials"
  ► Response Headers
  ▼ Response Body [🔗]
    access_token: "[REDACTED]"
    scope: "write:tradeitem compose:cic"
    expires_in: 86400
    token_type: "Bearer"
```

To upload trade items to the API, pass along the "access_token" received from the login request as a Bearer token using the Authorization header. In addition, your subscription key must be included as the value of the "Ocp-Apim-Subscription-Key" header.

<https://api-uat.gs1belu.org/myproductmanager/upload/v3/tradeitems>

```
POST https://api-uat.gs1belu.org/myproductmanager/upload/v3/tradeitems
  ► Network
  ▼ Request Headers
    Ocp-Apim-Subscription-Key: "[REDACTED]"
    Authorization: "Bearer [REDACTED]"
    access_token
  Content-Type: "application/json"
```

Note that, when you perform a POST operation against the API, you should also include the "Content-Type" header with value "application/json". This tells the API that you are supplying data in JSON format.

For security reasons we have implemented a limit on the number of calls our users can perform on the system.

- **Download**

- 10 calls/second (600/min window)
- 50k calls/day

- **Upload**

- 10 calls/second (600/min window)
- 10k calls/day

Uploading Data

The API accepts data formatted according to the Open API specification, which is available [here](#) in JSONv3 format.

The API will always respond with a “201 Created” response with an empty body, regardless of any errors that might exist in the data.

In the headers of the response, a “Location” header will be returned pointing to an address where the resulting trade item and its validation status can be retrieved:

▼ Response Headers

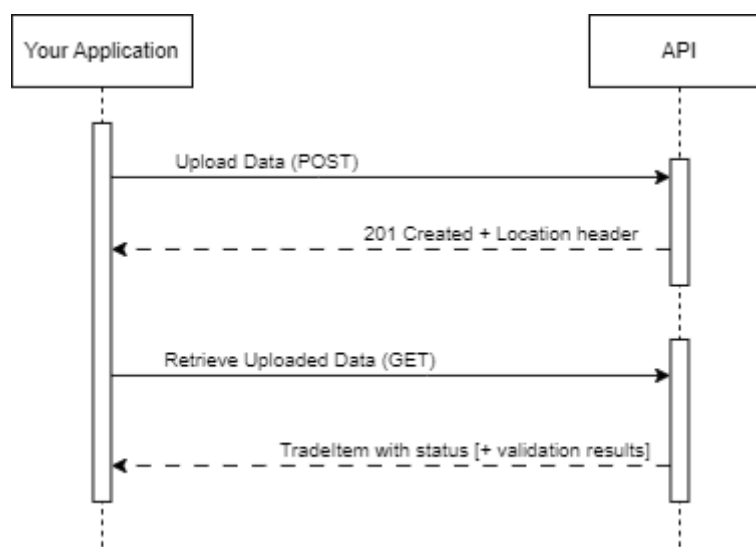
Content-Length: "27"

Content-Type: "application/json; charset=utf-8"

Location: "https://api-uat.gsibelu.org/myproductmanager/upload/v1/tradeitems/05410000000002"

Consulting Uploaded Data and Validation Results

The data uploaded through the API will be validated **asynchronously** after upload. To retrieve the validation progress and any errors that might exist, a **GET** request should be performed on the URI that was returned in the Location header in response to the upload request.



The API response will contain a "metaData" section with a status property:

```
"metaData": {
  "status": "active",
  "created": {
    "date": "2022-10-18T13:20:49.509Z"
  },
  "modified": {
    "date": "2022-10-18T13:22:24.415Z"
  },
  "isDiscontinued": false
},
```

While we are validating your upload, this status will be listed as **"pendingValidation"**. If this is the case, you should retry retrieving the data at a later time.

Once validated the status will change to either **"active"** or **"incomplete"**, depending on the result of our checks. In either case, the metaData section *might* contain a validationResults property containing any rules that were not satisfied in the uploaded data:

```
"metaData": {
  "validationResults": [
    {
      "severity": "error",
      "code": "VR_FMCGB2C_0257",
      "message": "No image is provided. Since 1/01/2021 every product should have at least one image (a packshot), unless the product will be discontinued before 1/09/2021."
    },
    {
      "severity": "warning",
      "code": "VR_FMCGB2C_0315",
      "message": "The field Consumer storage instructions must be filled out at least once, because the applied GPC code indicates that the product belongs to a group where such is required by law."
    }
  ],
  "status": "incomplete",
}
```

If any blocking rule was violated (severity "error"), then the status will be "incomplete". In this case, the uploaded data is deemed incorrect and is not considered for publication to data recipients.

Errors with severity "warning" are allowed for a certain time but are scheduled to be elevated to blocking after a certain grace period. If only warnings are present (or no rules are violated at all), the status will be "active", and your data will be published to recipients.

Versioning

This API will possibly change in the future. This API will follow Postel's Law: "Be conservative in what you do, be liberal in what you accept from others." This law has some implications for an API consumer and provider.

As a consumer, you should:

- Ignore new and unknown fields in the payload (cfr. tolerant reader pattern).
- Be prepared for new enumeration values. Provide default behaviour for unknown values, if possible.

As a provider, we will:

- Only add optional fields.
- Never change the meaning or type of a field.

The API will increase the version when the needed changes are no longer possible to implement in the current API without breaking changes. GS1 will try to support the old API version for three months before deprecation. This allows consumers time to start using the new API version.

All version increases will be communicated via mail to the registered mail address.

Publication settings

The supplier can determine which data recipients have access to their product data in My Product Manager. At this moment this setting can't be defined through the Upload API and the supplier should login to the web interface of My Product Manager to define the Publication settings. It is possible to set "Default" Publication settings that are taken into account for every new trade item that's created via the Upload API.

FAQ

1. Q: Why do I get the error {"error":"access_denied","error_description":"Non-global clients are not allowed access to APIv1"} when requesting a token?

A: Don't forget to add the audience in the request:
"audience": "https://api-uat.gs1belu.org/"

The trailing "/" at the end should be present!

2. Q: I uploaded data through the API, but it is not available to my data recipients?

A: Your publication settings may prevent us from publishing your data to one or more recipients. You can check these settings in the My Product Manager web interface.

If your publication settings are correct, your upload might have been rejected or is still being validated. Please check the status of your data using the GET endpoint. If an item has the status "incomplete", it will not be available to your configured recipients. If it has the status "pendingValidation", we are still processing your latest update, any data that was previously accepted should still be available to your recipients, however.